

Report Guidelines: Data Transformation to Handle Skewness in Machine Learning

Objective:

Prepare a concise report (4–10 pages) explaining skewness in numerical features, why it matters in machine learning, and how transformation techniques can reduce skewness.

1. Introduction

Define skewness and explain its importance in data preprocessing.

2. Understanding Skewness

Explain positive (right) and negative (left) skewness, include formulas if desired and illustrative figures.

3. Why Skewness Can Be Harmful

Discuss effects on statistical analysis, distance-based algorithms, linear models, outliers, and model performance.

4. Detecting Skewness

Explain visual methods (histogram, KDE, boxplot) and **numerical measures** (skewness coefficient).

5. Handling Positive Skewness

Explain log, square root, cube root, reciprocal (where appropriate), and Box-Cox transformation. State Box-Cox requires strictly positive values.

6. Handling Negative Skewness

Explain reflection plus transformation when appropriate and discuss Yeo-Johnson transformation for data containing zero or negative values.

7. Box-Cox vs. Yeo-Johnson

Compare assumptions, advantages, disadvantages, and suitable use cases in a table.

8. Practical Example

Apply Box-Cox and Yeo-Johnson on a skewed dataset. Show before/after histograms, boxplots, and skewness values.

9. Conclusion

Summarize key findings and recommendations.

10. Different Transformation Families

State different families of transformation. Mainly:

1. Function Transformers

2. Quantile Transformers

3. Power Transformers

When to use which? What is the common standard in real life ML applications?

Formatting Requirements

- Length: 4–10 pages.
- Use clear headings and figures.
- Cite all references (APA/IEEE).
- Include captions for all figures and tables.
- Submit as a PDF generated from your report.

Evaluation Rubric (20 Marks)

Criterion	Marks
Concept of skewness	3
Why skewness is problematic	3
Positive & negative skewness handling	4
Box-Cox explanation	3
Yeo-Johnson explanation	3
Comparison & practical example	2
Organization, references & formatting	2